

Problem Set 5: Difference-in-Differences

MY457 | WT 2026 | Week 10

Add your answers to this .qmd file by replacing the dedicated placeholder text. When you are finished, render the .qmd file as a .pdf and make sure you push your work to GitHub.

1 Concepts

- 1.1 In the table below, fill in the missing values and **bold** the values you fill in. Note that treatment is realised across two dimensions: treatment group $G_i \in 0, 1$, and time $t \in 0, 1$. The treatment indicator is thus given by Z_{it} . The potential outcomes are given by $Y_{it}(0)$ and $Y_{it}(1)$.

i	G_i	t_i	Z_{it}	$Y_{it}(0)$	$Y_{it}(1)$	Y_{it}	τ_{it}
1	0	0	0	5	10	?	?
2	0	1	?	6	11	?	?
3	1	0	0	7	9	?	?
4	?	?	1	8	10	?	?

 Answer:

i	G_i	t_i	Z_{it}	$Y_{it}(0)$	$Y_{it}(1)$	Y_{it}	τ_{it}
1	0	0	0	5	10	5	5
2	0	1	0	6	11	6	5
3	1	0	0	7	9	7	2
4	1	1	1	8	10	10	2

- 1.2 Explain the main identification assumption in difference-in-differences designs. Looking at the table above, can you establish whether this assumption holds? Why or why not?

 Answer

The main identification assumption in difference-in-differences designs is *parallel trends*. This assumption states that in the absence of treatment, the difference between the treated and control groups would have remained constant over time.

Formally:

$$E[Y_{i1}(0) - Y_{i0}(0)|G_i = 1] = E[Y_{i1}(0) - Y_{i0}(0)|G_i = 0]$$

If we plug in the values from the table of potential outcomes:

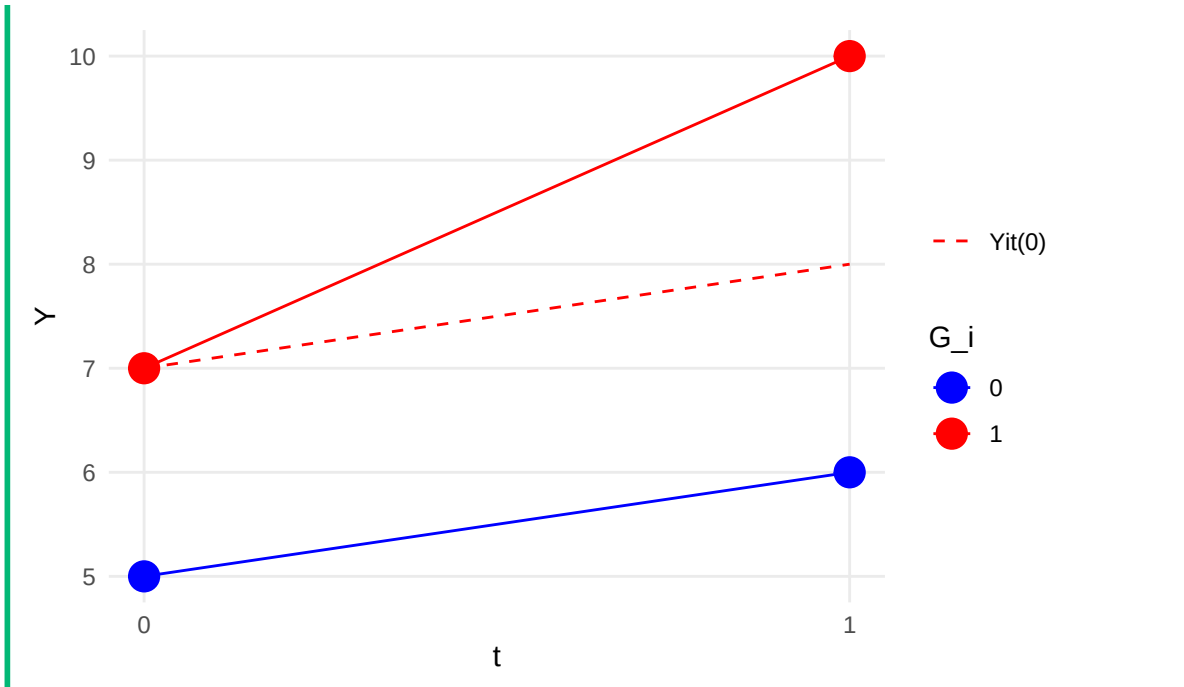
$$\begin{aligned}(8 - 7) &= (6 - 5) \\ 1 &= 1\end{aligned}$$

Hence, the parallel trends assumption holds in this case, since the change in potential outcomes under control is the same for both groups. We can also plot the data to visualise this assumption:

```

tibble(
  G = c(0, 0, 1, 1),
  t = c(0, 1, 0, 1),
  Z = c(0, 0, 0, 1),
  Y_0 = c(5, 6, 7, 8),
  Y_1 = c(10, 11, 9, 10),
  Y = c(5, 6, 7, 10),
) |>
  ggplot(aes(x = t, y = Y, color = factor(G))) +
  geom_point(size = 5) +
  geom_line() +
  # add potential outcomes
  geom_line(
    data = \ (d) filter(d, G == 1),
    aes(
      y = Y_0,
      linetype = "Yit(0)"
    ),
    color = "red"
  ) +
  labs(color = "G_i", linetype = "") +
  scale_color_manual(
    values = c("blue", "red"),
    labels = c("0", "1")
  ) +
  scale_linetype_manual(values = c("dashed")) +
  scale_x_continuous(
    "t",
    breaks = c(0, 1),
    labels = c("0", "1"),
    minor_breaks = NULL
  ) +
  scale_y_continuous(minor_breaks = NULL) +
  theme_minimal()

```



1.3 Write down the typical 2-period difference-in-differences estimand using potential outcomes notation. Calculate this quantity using the table provided.

💡 Answer

The estimand for τ_{ATT} is given by:

$$\begin{aligned}\tau_{ATT} &= E[Y_{i1}(1) - Y_{i1}(0) | G_i = 1] \\ &= E[Y_{i1}(1) | G_i = 1] - E[Y_{i1}(0) | G_i = 1]\end{aligned}$$

Hence, we can calculate the true ATT using the potential outcomes in the table:

$$\begin{aligned}\tau_{ATT} &= E[Y_{i1}(1) | G_i = 1] - E[Y_{i1}(0) | G_i = 1] \\ &= 10 - 8 \\ &= 2\end{aligned}$$

- 1.4 Write down the plug-in estimator for the ATT in the simple two-period, two-group difference-in-differences design. Estimate the ATT using the observed data in the table above.

 Answer

The estimator for the ATT in the simple two-period, two-group difference-in-differences design is given by:

$$\begin{aligned}\hat{\tau}_{ATT} &= [E[Y_{i1}|G_i = 1] - E[Y_{i0}|G_i = 1]] \\ &\quad - [E[Y_{i1}|G_i = 0] - E[Y_{i0}|G_i = 0]]\end{aligned}$$

Using the observed data in the table above, we can estimate the ATT as follows:

$$\begin{aligned}\hat{\tau}_{ATT} &= (10 - 7) - (6 - 5) \\ &= 3 - 1 \\ &= 2\end{aligned}$$

Note that swapping the dimensions (ie. estimating the difference in the difference between groups instead of the difference in the difference between time periods) would give us the same estimate for the ATT:

$$\begin{aligned}\hat{\tau}_{ATT} &= [E[Y_{i1}|G_i = 1] - E[Y_{i1}|G_i = 0]] \\ &\quad - [E[Y_{i0}|G_i = 1] - E[Y_{i0}|G_i = 0]] \\ &= (10 - 6) - (7 - 5) \\ &= 4 - 2 \\ &= 2\end{aligned}$$

- 1.5 Scrutinise the two estimators below. When would we use one over the other?

1.

$$Y_{it}(Z) = \alpha_i + \gamma t + \tau Z + \epsilon_{it}$$

2.

$$\hat{Y}_i = \hat{\mu} + \hat{\gamma}G_i + \hat{\delta}T + \hat{\tau}G_i t + \hat{\epsilon}_i$$

Answer

The first estimator is a fixed effects regression where we control for unit fixed effects α_i and time fixed effects γt . We use this estimator when we have panel data, where we observe the same units across time. The second estimator is a simple OLS regression where we control for group fixed effects $\hat{\gamma}G_i$ and time fixed effects $\hat{\delta}T$. We use this estimator when we have repeated cross-sectional data, where we observe different units across time.

2 Simulations

2.1 Explain the code below and relate it to a difference-in-differences data generating process. What kind of data (panel or repeated-cross sectional) is this?

```
set.seed(123)
n_units <- 1000
tau <- 25000

G <- rbinom(n_units, 1, 0.5)
for (i in 1:2) {
  data <- tibble(
    ID = 1:n_units,
    G = G,
    T = ifelse(i == 2, 1, 0)
  )

  if (i == 1) {
    sim_data <- data
  } else {
    sim_data <- rbind(sim_data, data)
  }
}

Y0 <- rnorm(n_units, 50000, 2500)

data1 <- sim_data |>
  mutate(
    Y0 = c(Y0, Y0 * 1.1),
    Y0 = ifelse(G == 1, Y0 + 10000, Y0),
    Y1 = Y0 + tau,
```

```
Y = ifelse(G == 1 & T == 1, Y1, Y0)
)
```

💡 Answer

We are creating a data frame with 1000 units. We randomly assign units to treatment and control groups (G) and specify two periods, one pre-treatment and one post-treatment (T).

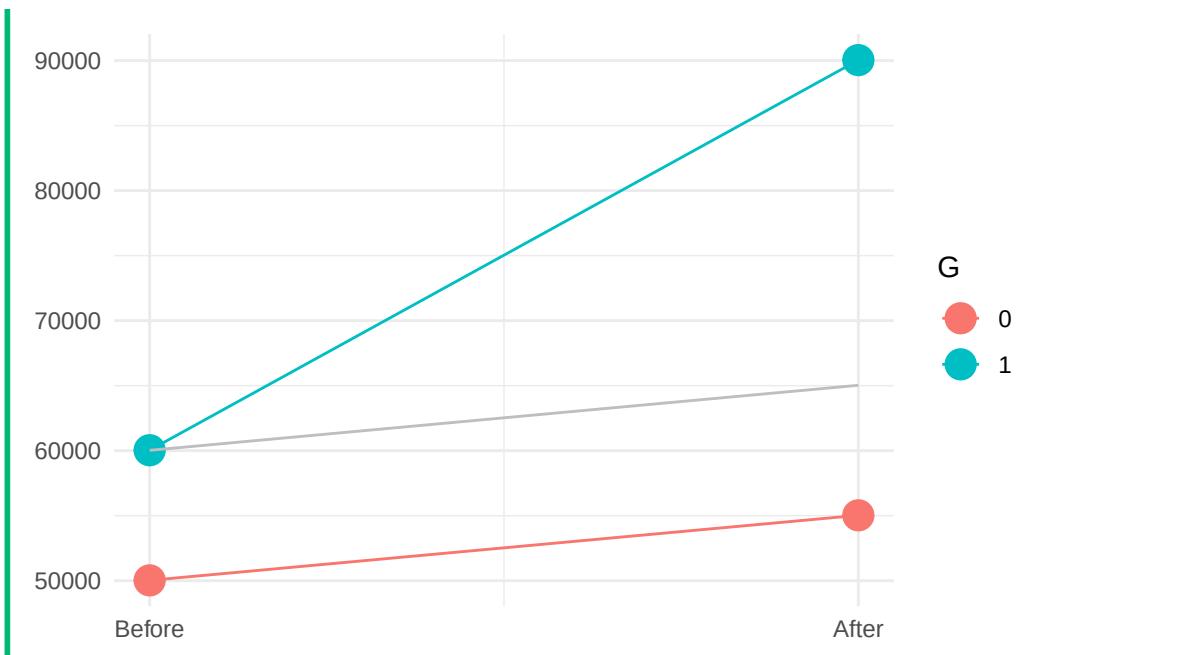
We create a continuous potential outcome under control which is normally distributed. In the post-period, we increase this by 10%. We define our treatment effect to be 25,000 and add it to the potential outcome under control.

Lastly, we define our realized outcome based on treatment group and treatment period (G and T).

2.2 Using the potential outcomes in our simulated data, create a plot visualizing the difference-in-differences estimator.

💡 Answer

```
data1 |>
  group_by(G, T) |>
  summarise(Y = mean(Y)) |>
  ggplot() +
  aes(x = T, y = Y, color = factor(G)) +
  geom_point(size = 5) +
  geom_line() +
  geom_line(data = \d filter(d, G == 0), aes(y = Y + 10000), color = "grey") +
  labs(color = "G") +
  scale_x_continuous(breaks = c(0, 1), labels = c("Before", "After")) +
  theme_minimal() +
  xlab("") +
  ylab("")
```



2.3 Without using a regression, estimate the canonical two-period difference-in-differences using only Y , G , and t . What do you find?

💡 Answer

```
data_means <- data1 |>
  group_by(G, T) |>
  summarise(Y = mean(Y))

# estimate DiD
with(data_means, {
  # pre-post difference for treated
  (Y[G == 1 & T == 1] - Y[G == 1 & T == 0]) -
  # pre-post difference for control
  (Y[G == 0 & T == 1] - Y[G == 0 & T == 0])
})
```

```
[1] 25000.84
```

```
# again, this is the same as:
with(data_means, {
  # treated-control difference in post period
  (Y[G == 1 & T == 1] - Y[G == 0 & T == 1]) -
  # treated-control difference in pre period
  (Y[G == 1 & T == 0] - Y[G == 0 & T == 0])
})
```

```
[1] 25000.84
```

When estimating the canonical Diff-in-Diff, we find an estimated treatment effect of 25,000.84

2.4 Now estimate the difference-in-differences design using linear regression. Do you find any differences to your previous estimation? Why or why not?

 Answer

```
lm(Y ~ factor(G) * factor(T), data = data1) |>
  modelsummary(
    stars = T,
    gof_map = c("nobs" = "N"),
    output = "markdown"
  )
```

	(1)
(Intercept)	50025.715*** (116.948)
factor(G)1	10008.352*** (166.559)
factor(T)1	5002.571*** (165.389)
factor(G)1 × factor(T)1	25000.835*** (235.550)

The estimated treatment effect is 25,000.84, which is the same as the previous estimation.

2.5 Now consider a new data generating process, given by the simulation code below.
Explain how this data generating process differs from the previous one.

```
set.seed(123)

n_units <- 1000
n_periods <- 20
att_0 <- 5000

units <- tibble(
  ID = 1:n_units,
  G = rbinom(n_units, 1, 0.5),
  Y0_base = rnorm(n_units, 50000, 2500),
  cohort = if_else(G == 1L, sample(6:n_periods, n_units, replace = TRUE), NA_integer_)
)

data2 <- units |>
  cross_join(tibble(P = 1:n_periods)) |>
  mutate(
    event_time = if_else(!is.na(cohort), P - cohort, NA_integer_),
    Z = if_else(!is.na(event_time) & event_time >= 0, 1L, 0L),
    tau_it = if_else(Z == 1L, att_0 + 500 * event_time, 0),
    Y0 = Y0_base * (1 + P / 10) + G * 10000,
    Y1 = Y0 + tau_it,
    Y = if_else(Z == 1L, Y1, Y0)
  )
```

Answer

Firstly, the first DGP is a classic 2x2 design (one pre, one post). Here we have 20 periods, with at least 5 pure pre-treatment periods for everyone.

In the first DGP, all treated units ($G = 1$) receive treatment at the same time (period 1). Here, each treated unit has its own adoption period drawn uniformly from periods 6–20, so treatment is *staggered*.

Finally, the first DGP has a constant effect $\tau = 25,000$ that is the same for every treated unit in every period. Here, the effect depends on exposure: $\tau_{it} = 5000 + 500 \times \text{event_time}$, so it grows the longer a unit has been treated. A unit adopted in period 6 has a much larger effect by period 20 than one adopted in period 18.

Variable	Description
ID	Unit identifier (1 to 1,000)
G	Treatment group indicator (1 = eventually treated, 0 = never treated)
Y0_base	Unit-level baseline earnings, drawn once and fixed across periods
cohort	Period in which treatment begins (6–20 for treated units, NA for control)
P	Time period (1 to 20)
event_time	Periods since treatment adoption ($P - \text{cohort}$), NA for control units
Z	Treatment status indicator (1 = currently treated, 0 = not yet or never treated)
tau_it	Unit-period treatment effect ($5000 + 500 \times \text{event_time}$)
Y0	Untreated potential outcome, with common time trend and group-level shift
Y1	Treated potential outcome ($Y_0 + \tau_{it}$)
Y	Observed outcome (switches from Y_0 to Y_1 upon adoption)

2.6 Using the new simulated data, estimate the difference-in-differences design using a two-way fixed effects linear regression. You can do this in multiple ways: using `lm` and `factor()`, using `plm` with `model = "within"` and `effect = "twoways"`, or using `fixest`. Pick two of these methods and compare the results in a table (omitting FEs from the output).

 Tip

Answer

```

sdid_lm <- lm(Y ~ Z + factor(P) + factor(ID), data = data2)

sdid_plm <- plm(Y ~ Z,
  data = data2,
  index = c("ID", "P"),
  model = "within", effect = "twoways"
)

sdid_feols <- feols(Y ~ Z | ID + P, data = data2)

modelsummary(
  list(
    "LM" = sdid_lm,
    "PLM" = sdid_plm,
    "FEOLS" = sdid_feols
  ),
  stars = T,
  gof_omit = "AIC|BIC|RMSE",
  output = "markdown",
  coef_map = c("Z" = "Treatment Effect (Z)")
)

```

	LM	PLM	FEOLS
Treatment Effect (Z)	6479.438*** (44.733)	6479.438*** (44.733)	6479.438*** (129.873)
Num.Obs.	20000	20000	20000
R2	0.997	0.525	0.997
R2 Adj.	0.997	0.500	0.997
R2 Within			0.525
R2 Within Adj.			0.525
Log.Lik.	-175899.568		
F	7176.141		
Std.Errors			by: ID
FE: ID			X
FE: P			X

Note that we get the same estimate for all three methods. However, `feols` by default clusters standard errors by the first fixed effect (meaning the first variable behind `|`, in this case `ID`), while `lm` and `plm` do not cluster standard errors. The other methods assume independent and identically distributed errors, which is not the case in panel data. As treatment is assigned at the unit level, clustering by `ID` is appropriate.

2.7 Using `data2` and either the `fect` package or the `did` package, estimate dynamic period-specific ATTs and provide an event study plot. What do you find?

Answer

```
# using `fect`  
out.fect <- fect(Y ~ Z,  
  data = data2, index = c("ID", "P"),  
  method = "fe", force = "two-way", se = TRUE, nboots = 200  
)  
  
print(out.fect)
```

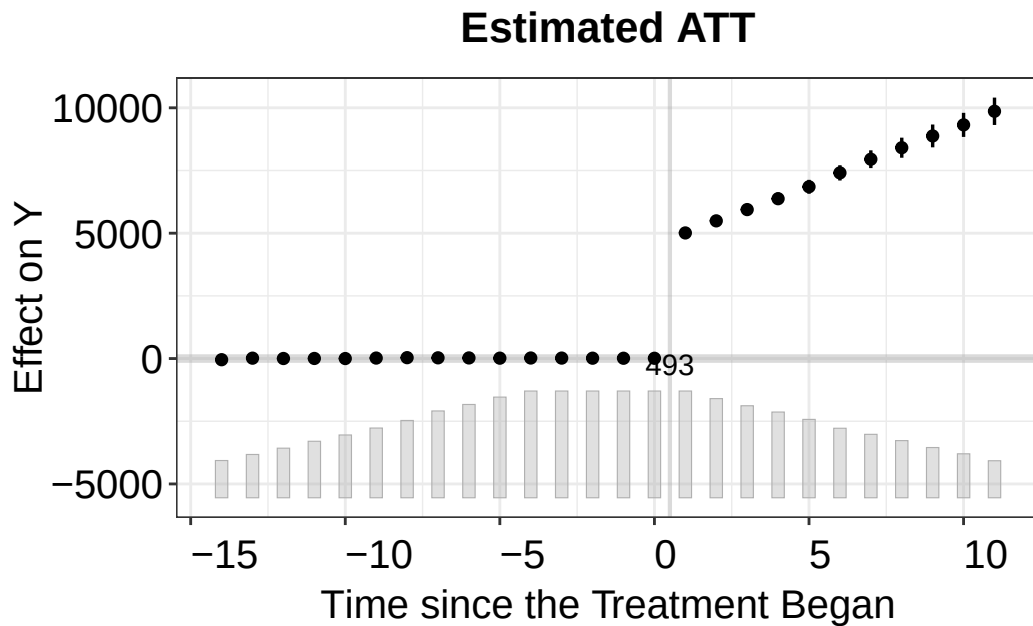
Call:

```
fect.formula(formula = Y ~ Z, data = data2, index = c("ID", "P"),  
  force = "two-way", method = "fe", se = TRUE, nboots = 200)
```

ATT:

	ATT	S.E.	CI.lower	CI.upper	p.value
Tr obs equally weighted	7251	144.2	6969	7534	0
Tr units equally weighted	6727	135.3	6462	6992	0

```
plot(out.fect)
```



```
# using `did`
data2_did <- data2 |>
  mutate(
    ID = ID,
    cohort = ifelse(is.na(cohort), 0, cohort)
  )

out.did <- att_gt(
  yname = "Y",
  tname = "P",
  idname = "ID",
  gname = "cohort",
  data = data2_did,
  panel = TRUE
)

did::aggte(out.did, type = "simple")
```

Call:
 did::aggte(MP = out.did, type = "simple")

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Mul

ATT	Std. Error	[95% Conf. Int.]
7297.32	113.0239	7075.797 7518.842 *

Signif. codes: `*' confidence band does not cover 0

Control Group: Never Treated, Anticipation Periods: 0

Estimation Method: Doubly Robust

The `fect` function reports two estimates for the ATT, one weighting each observation equally, one weighting each unit equally (note that units appear multiple times in panel data). For the unit-weighted ATT, the function computes each treated unit's mean effect across all its treated periods, then average those. This means that the treatment effect in each cohort contributes proportional to the share of units that are first treated in that cohort, regardless of how many treated periods that cohort has. The observation-weighted ATT gives more weight to cohorts with more treated periods, since those cohorts contribute more observations to the data.

The `did` package requires us to recode cohorts that were never treated to 0. The `att_gt` function estimates the group-time average treatment effects, only using the "good" control groups (never-treated and/or not-yet-treated). We can manually pick which group to use with the `control_group` argument but the default `nevertreated` is usually reasonable. To make sense of the many group-time ATTs, we can aggregate them using the `aggte` function. The `type` argument specifies how we aggregate the ATTs. The `simple` type gives us a single overall ATT, which is a weighted average of the group-time ATTs. As you can see, it closely mirrors the observation-weighted ATT from `fect`.

We can also use the `aggte` function to get dynamic period-specific ATTs, which are the average treatment effects for each event time (relative to adoption). The `dynamic` type gives us these period-specific ATTs, which we can plot using the `ggdid` function. The `group` type gives us the average treatment effect for each cohort (group), which we can also plot using `ggdid`.

```
did::aggte(out.did, type = "dynamic")
```

Call:

```
did::aggte(MP = out.did, type = "dynamic")
```

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Mul

Overall summary of ATT's based on event-study/dynamic aggregation:

ATT	Std. Error	[95% Conf. Int.]
8420.752	186.9584	8054.32 8787.183 *

Dynamic Effects:

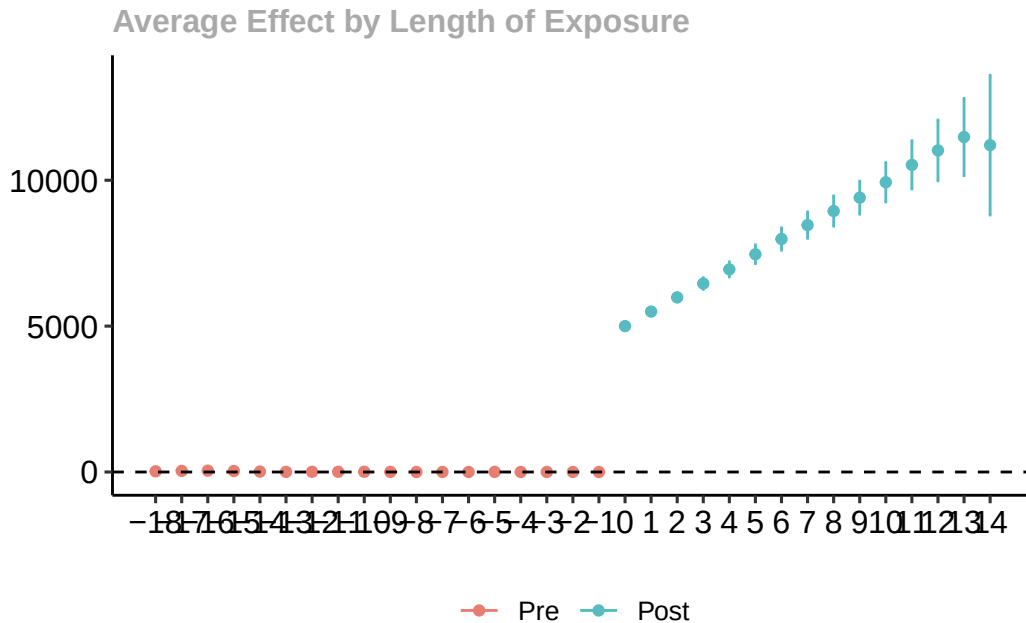
Event time	Estimate	Std. Error	[95% Simult. Conf. Band]
-18	22.6680	45.5647	-101.4685 146.8045
-17	37.6651	29.0877	-41.5815 116.9117
-16	45.5452	27.9339	-30.5580 121.6484
-15	33.7551	25.4473	-35.5736 103.0838
-14	13.6034	21.0801	-43.8272 71.0341
-13	5.0314	21.5438	-53.6627 63.7254
-12	7.0913	19.4543	-45.9100 60.0926
-11	7.1136	19.6473	-46.4134 60.6406
-10	8.1666	19.3254	-44.4835 60.8167
-9	4.6395	18.1510	-44.8111 54.0900
-8	0.3379	17.5128	-47.3739 48.0497
-7	0.6364	15.9032	-42.6905 43.9632
-6	1.1569	16.1093	-42.7314 45.0451
-5	4.0812	15.9150	-39.2775 47.4399
-4	0.8352	15.8075	-42.2307 43.9012
-3	0.8352	15.4719	-41.3166 42.9871
-2	0.8352	15.5100	-41.4202 43.0907
-1	0.8352	16.7301	-44.7444 46.4149
0	5000.8352	15.2302	4959.3420 5042.3284 *
1	5498.3336	32.0605	5410.9880 5585.6792 *
2	5984.8274	50.3437	5847.6709 6121.9839 *
3	6459.5342	63.4129	6286.7721 6632.2963 *
4	6944.6113	79.1293	6729.0312 7160.1914 *
5	7463.9623	106.0327	7175.0866 7752.8381 *
6	7985.7969	124.8360	7645.6935 8325.9004 *
7	8463.2685	146.8091	8063.3014 8863.2355 *
8	8943.9488	185.5945	8438.3147 9449.5828 *
9	9403.6186	205.2577	8844.4140 9962.8231 *
10	9930.3891	238.5208	9280.5625 10580.2158 *
11	10525.6889	304.4018	9696.3762 11355.0017 *
12	11022.1274	350.7681	10066.4941 11977.7608 *
13	11480.3933	436.9764	10289.8938 12670.8928 *
14	11203.9383	808.1089	9002.3247 13405.5518 *

Signif. codes: `*' confidence band does not cover 0

Control Group: Never Treated, Anticipation Periods: 0

Estimation Method: Doubly Robust

```
ggdid(did::aggte(out.did, type = "dynamic"))
```



The dynamic aggregation shows that the treatment effect increases over time. Units that have been exposed to the treatment for only one period have an ATT of 5000 (recall that this is the `att_0` parameter in our DGP), while units with longer exposure have higher ATTs. Consistent with the way we defined τ_{it} in our DGP, the ATT increases by ≈ 500 for each additional period of exposure. At the same time, the dynamic aggregation shows that we don't observe any pre-trends, which strengthens the plausibility of the parallel trends assumption.

Finally, we can also look at the group-specific ATTs, which show that earlier cohorts have higher ATTs in later periods, consistent with the fact that they have been exposed to treatment for longer.

```
did::aggte(out.did, type = "group")
```

Call:

```
did::aggte(MP = out.did, type = "group")
```

Reference: Callaway, Brantly and Pedro H.C. Sant'Anna. "Difference-in-Differences with Mul

Overall summary of ATT's based on group/cohort aggregation:

ATT	Std. Error	[95% Conf. Int.]
6726.605	71.106	6587.24 6865.971 *

Group Effects:

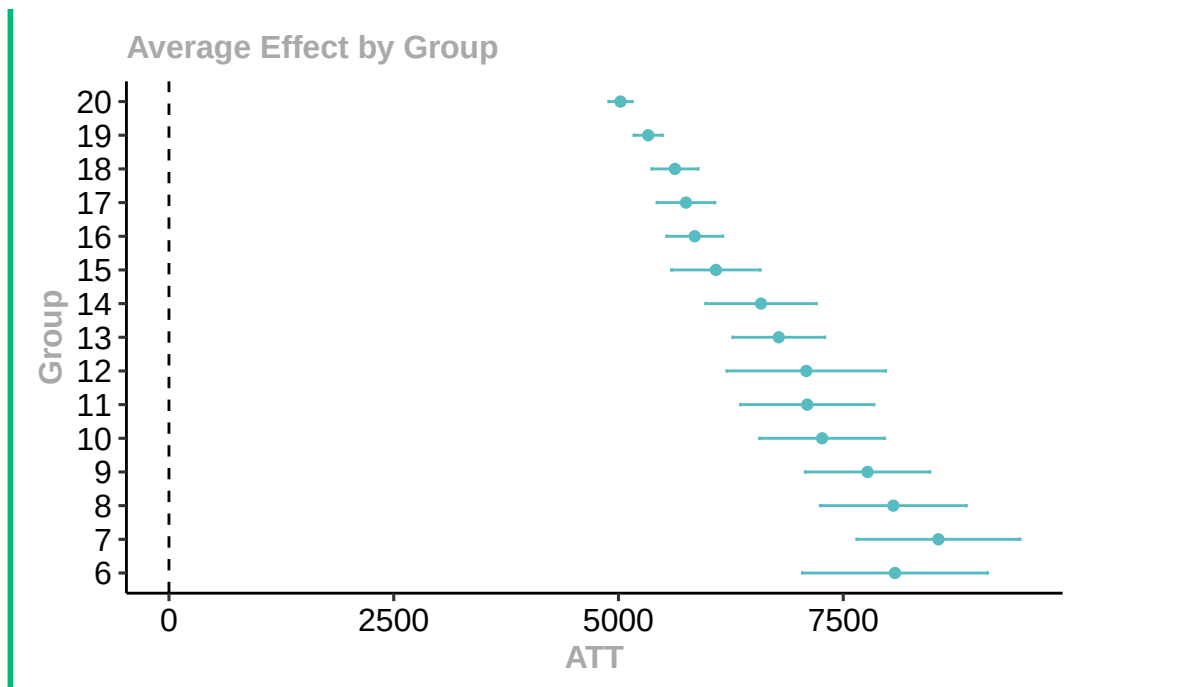
Group	Estimate	Std. Error	[95% Simult. Conf. Band]
6	8075.434	403.0843	6973.011 9177.856 *
7	8558.637	302.1527	7732.259 9385.014 *
8	8056.799	305.2408	7221.975 8891.623 *
9	7769.876	254.3985	7074.105 8465.648 *
10	7264.582	243.2105	6599.409 7929.754 *
11	7099.711	242.6609	6436.041 7763.380 *
12	7088.218	327.6376	6192.140 7984.296 *
13	6782.728	172.7214	6310.341 7255.116 *
14	6585.192	213.1654	6002.191 7168.192 *
15	6083.310	168.0043	5623.824 6542.797 *
16	5847.649	112.2657	5540.606 6154.692 *
17	5750.296	128.9027	5397.752 6102.841 *
18	5628.045	91.5729	5377.596 5878.494 *
19	5330.357	57.3535	5173.497 5487.217 *
20	5022.668	43.2130	4904.482 5140.854 *

Signif. codes: '*' confidence band does not cover 0

Control Group: Never Treated, Anticipation Periods: 0

Estimation Method: Doubly Robust

```
ggdid(did::aggte(out.did, type = "group"))
```



3 Replication

In this section, we will use real-world data to reinforce what we have learned. We will analyse the dataset employed in *The Effects of Income Transparency on Well-Being: Evidence from a Natural Experiment* (Perez-Truglia, 2020).

In recent decades, there has been an increasing push towards higher transparency in income, wealth, and earnings. Transparency facilitates comparisons between individuals. In 2001, Norwegian tax records became accessible online allowing individuals to have access to these easily, assuming they had access to internet.

The author uses this setting to analyse the effect of salary transparency on the subjective well-being of individuals across the income distribution.

3.1 Skim the paper and fill out the table below to provide a summary of the paper's research design (focus on equation 1 of the paper). Be as precise and concise as possible. Edit the table directly in your .qmd to provide your answers.

Category	Description
Research Question	
Unit of Analysis	

Category	Description
Data structure	
t	
G	
Y	
Control Group	

💡 Answer

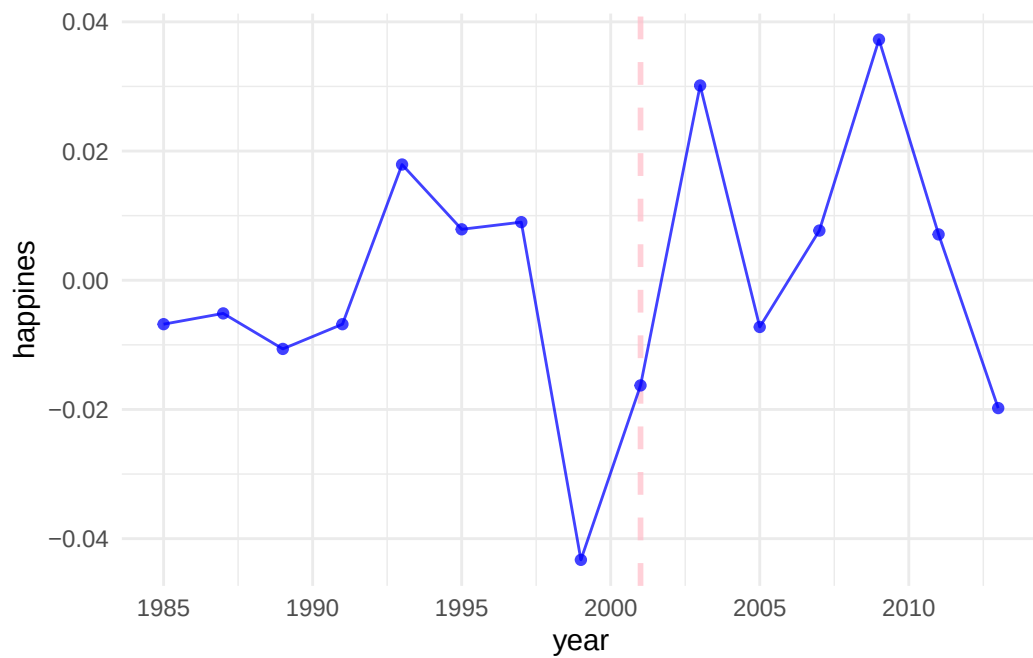
Category	Description
Research Question	Does income transparency increase the happiness gap between rich and poor?
Unit of Analysis	Survey Respondents
Data structure	Repeated Cross Section
t	$I_t^{01-13} \in \{0, 1\}$ a dummy for the time period 2001 - 2013, when transparency was high, 0 for years before, when income transparency was low.
G	IncomeRank $_{i,t}$ a continuous variable for respondent i 's income rank at time t . It is normalised to run from 0 to 1 so that $t \in [0, 1]$
Y	Subjective well-being
Control Group	Trick question: because everyone is treated (can see income information) there isn't a real control group, certainly a weakness of the paper. Instead, the authors test whether the association between income rank and subjective well-being grows after treatment.

3.2 Read into R the replication data set (`Norway-MSD.dta`) and visualise the trend in Norwegian happiness (`po_happy`) over the years. Include a vertical line to indicate when treatment came into effect.

💡 Answer

```
df <- read_dta(here("2026", "did", "data", "Norway-MSD.dta"))

df |>
  group_by(year) |>
  summarise(
    happines = mean(po_happy, na.rm = T)
  ) |>
  ggplot() +
  aes(x = year, y = happines) +
  geom_vline(xintercept = 2001, color = "pink", linetype = "dashed", alpha = 0.75, size = 2) +
  geom_point(color = "blue", alpha = 0.75) +
  geom_line(color = "blue", alpha = 0.75) +
  theme_minimal()
```



- 3.3 Why can the author just compare the dependent variable before and after treatment or between groups in the post-treatment period? Which types of confounders does the design eliminate and which ones not?

 Answer

If they compared the treated group before and after (possibly separately for each level of IncomeRank), any variable that changes over time would qualify as a possible confounder. If the subjective well-being in Norway increased over the course of the study, for example because of a flourishing economy, this would induce bias. By controlling for time fixed effects, the author eliminates this source of confounding.

If the author compared individuals on the bottom and the top of the income distribution in the post period, all variables that could potentially affect income and the dependent variable would be plausible confounders. By controlling for income rank, the authors close all backdoor paths ($\text{IncomeRank}_{i,t} \times I_t^{01-13} \leftarrow \text{IncomeRank}_{i,t} \leftarrow U \rightarrow Y$).

However, the design doesn't eliminate time-varying confounders. If, for instance, the oil price affects individuals at different points of $\text{IncomeRank}_{i,t}$ differently and the oil price changed over time, this would induce bias.

- 3.4 Estimate the baseline specification as given in equation (1) in the paper using `fixest::feols()`. In addition to the difference-in-differences components, the regression should include a dummy variable for each year, and control for each level of marital status, education, household size, household workers, and gender (`female`). Additionally, age and age squared should be included as continuous control variables. Use heteroscedasticity robust standard errors. What do you find?

 Answer

```
feols(
  po_happy ~ imp_hh_rank * post_2001 + poly(age, 2) |
    marital_status + education
    + hh_size + hh_workers + female + year,
  data = df,
  vcov = "hetero"
) |> modelsummary(
  stars = T,
  gof_omit = "AIC|BIC|RMSE",
  output = "markdown"
)
```

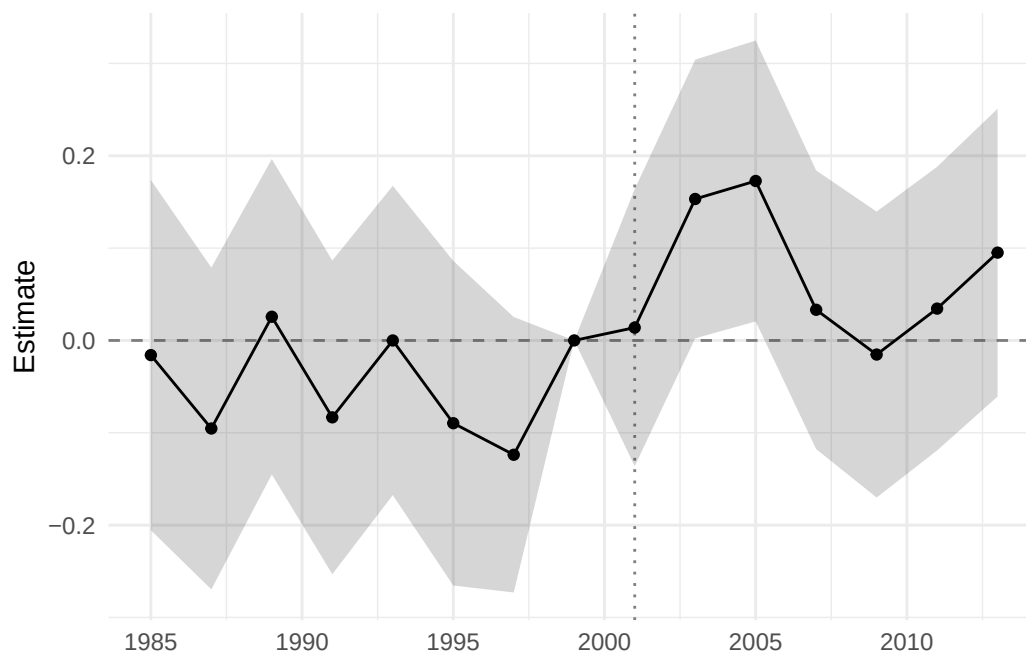
	(1)
imp_hh_rank	0.282*** (0.027)
poly(age, 2)1	-30.164*** (1.630)
poly(age, 2)2	27.489*** (1.311)
imp_hh_rank × post_2001	0.121*** (0.032)
Num.Obs.	46789
R2	0.071
R2 Adj.	0.070
R2 Within	0.022
R2 Within Adj.	0.022
Std.Errors	Heteroskedasticity-robust
FE: marital_status	X
FE: education	X
FE: hh_size	X
FE: hh_workers	X
FE: female	X
FE: year	X

`feols` converts all variables in the fixed effects part of the formula (after `|`) into factors, so we don't need to manually convert them. Note that we don't want to convert age and age squared into factors and hence included them in the first part of the formula. The coefficient for `imp_hh_rank x post_2001` is 0.121, meaning that the association between income rank and happiness increased by 0.121 points after treatment.

- 3.5 Using the same control variables as above, create an event study plot to test for parallel pre-trends. What do you find? Is this proof the parallel trends assumption holds/is violated? Why or why not?

💡 Answer

```
lm(po_happy ~ imp_hh_rank * relevel(factor(year), ref = "1999")
  + factor(marital_status) +
  factor(education) + factor(hh_size) + factor(hh_workers)
  + factor(female) + poly(age, 2), df) |>
tidy(conf.int = TRUE, conf.level = 0.95) |>
filter(str_detect(term, ":")) |>
mutate(year = as.integer(str_sub(term, -4, -1))) |>
add_row(year = 1999, estimate = 0, conf.low = 0, conf.high = 0) |>
ggplot(aes(x = year, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_vline(xintercept = 2001, linetype = "dotted", color = "grey50") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) +
  geom_line() +
  geom_point() +
  labs(x = NULL, y = "Estimate") +
  theme_minimal()
```



None of the coefficients in the pre-treatment period are statistically significant as shown by the confidence intervals. Significant pre-trends would be a red flag casting doubt on the validity of the parallel trends assumption. However, the absence of significant pre-trends is not proof that the parallel trends assumption holds. The parallel trends assumption is fundamentally untestable, since we can never observe the counterfactual

trend for the treated group.

- 3.6 The author argues that we should expect stronger effects among individuals with better access to the internet. Test their claim. Do the findings make you more or less confident that their interpretation of the results is correct?

💡 Answer

```
feols(  
  po_happy ~ imp_hh_rank * post_2001 * higher_internet + poly(age, 2) |  
    marital_status + education + hh_size + hh_workers + female + year,  
  data = df,  
  vcov = "hetero"  
) |> modelsummary(  
  stars = T,  
  gof_omit = "AIC|BIC|RMSE",  
  output = "markdown"  
)
```

	(1)
imp_hh_rank	0.288*** (0.040)
higher_internet	0.022 (0.034)
poly(age, 2)1	-30.526*** (1.822)
poly(age, 2)2	27.459*** (1.317)
imp_hh_rank × post_2001	0.027 (0.052)
imp_hh_rank × higher_internet	-0.038 (0.054)
post_2001 × higher_internet	-0.120** (0.043)
imp_hh_rank × post_2001 × higher_internet	0.209** (0.074)
Num.Obs.	46789
R2	0.071
R2 Adj.	0.070

R2 Within	0.022
R2 Within Adj.	0.022
Std.Errors	Heteroskedasticity-robust
FE: marital_status	X
FE: education	X
FE: hh_size	X
FE: hh_workers	X
FE: female	X
FE: year	X

The three way interaction between `imp_hh_rank`, `post_2001`, and `higher_internet` is positive and statistically significant, meaning that the increase in the association between income rank and happiness after treatment was stronger for individuals with better access to the internet. This finding is consistent with the author's interpretation of the results, since better access to the internet likely facilitated access to income information.

3.7 All in all, are you convinced by the author's research design and interpretation of the results? Why or why not?

Answer

There are several reasons to be sceptical of the research design:

1. No clear control group - this makes it difficult to derive a clear causal interpretation from the findings.
2. Given the design, anything that happened between 1999 and 2001 and affected people at different points of the income distribution differently, could be driving the results.
3. The parallel pre-trends seem to hinge on the conditioning set. If we recreate the event study plot without controls, it looks like the high values for 2003 and 2005 are merely a continuation of a trend that started in 1997:

```
lm(po_happy ~ imp_hh_rank * relevel(factor(year), ref = "1999"), df) |>
  tidy(conf.int = TRUE, conf.level = 0.95) |>
  filter(str_detect(term, ":")) |>
  mutate(year = as.integer(str_sub(term, -4, -1))) |>
  add_row(year = 1999, estimate = 0, conf.low = 0, conf.high = 0) |>
  ggplot(aes(x = year, y = estimate)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_vline(xintercept = 2001, linetype = "dotted", color = "grey50") +
  geom_ribbon(aes(ymin = conf.low, ymax = conf.high), alpha = 0.2) +
  geom_line() +
  geom_point() +
  labs(x = NULL, y = "Estimate") +
  theme_minimal()
```

